Robust Statistics: An Overview

2026-04-17

Introduction

A robust statistic is one that remains useful when the data deviate from the idealised model – heavy tails, outliers, skewed contamination. Classical estimators like the mean and SD are efficient under normality but break down under even modest contamination. Robust alternatives sacrifice a little efficiency for dramatic stability, and are the default in many applied fields where data quality cannot be guaranteed.

Prerequisites

The reader should know the mean, median, SD, and MAD. Familiarity with outlier concepts helps.

Theory

Two key concepts from robustness theory:

Breakdown point. The largest fraction of contaminated observations an estimator can tolerate before giving an arbitrary answer. The arithmetic mean has breakdown 0: a single extreme point can push it arbitrarily far. The median has breakdown 0.5: up to half the sample can be contaminated before the median is affected.

Influence function. Measures the infinitesimal effect of adding a single observation at x on the estimator. The arithmetic mean has a linear, unbounded influence function; the median has a bounded (step-shaped) influence function.

Robust estimators aim for high breakdown point and bounded influence function, typically with moderate loss of efficiency at the normal model.

Robust alternatives to classical summaries:

Classical	Robust	Breakdown	Efficiency at normal
Mean	Median	0.5	64%
Mean	20% trimmed mean	0.2	~95%
Mean	Huber M-estimator	0	~95%
SD	MAD	0.5	37%
SD	Qn, Sn	0.5	~82%
OLS	Robust regression (`rlm`, `lmrob`)	0 to 0.5	95%

Assumptions

Robustness is an explicit relaxation of distributional assumptions; it asks how the estimator behaves under deviation, not whether deviation is present.

R Implementation

```
library(robustbase)
library(MASS)
```

```r
set.seed(2026)
clean <- rnorm(100, mean = 50, sd = 10)
contaminated <- c(clean[1:95], rnorm(5, mean = 200, sd = 5))

classical <- c(mean = mean(contaminated), sd = sd(contaminated))
robust    <- c(median = median(contaminated),
               mad    = mad(contaminated),
               Qn     = Qn(contaminated),
               Sn     = Sn(contaminated))

rbind(classical = c(classical, NA, NA), robust)

# Robust regression example
n <- 80
x <- rnorm(n)
y <- 2 + 1.5 * x + rnorm(n)
y[c(5, 10, 15)] <- y[c(5, 10, 15)] + c(15, -15, 20)  # inject outliers

coef(lm(y ~ x))
coef(rlm(y ~ x))          # M-estimator
coef(lmrob(y ~ x))        # MM-estimator with high breakdown
```

Output & Results

```
      mean    sd    median  mad    Qn     Sn
 57.84   32.9   50.28   10.21  10.45 10.72
```

Classical mean is pulled to 57.8 by the five extreme values; every robust estimator remains near the true 50.

For the regression: OLS slope is 1.23 (pulled by outliers); `rlm` gives 1.47 (close to truth of 1.5); `lmrob` gives 1.48.

Interpretation

Use robust statistics as the default in biomedical work where outliers or heavy tails are typical. A robust summary plus a classical one can be reported side by side; divergence between them signals contamination that warrants investigation.

Practical Tips

- MM-estimators (`lmrob`) combine high breakdown with high efficiency and are the modern robust-regression default.
- Robust methods are not a license to ignore outliers; always investigate before reporting.
- For heavy-tailed data, a Student-t likelihood (`brms` or `heavy` package) is an alternative to robust M-estimators.
- Robust confidence intervals are usually narrower than a non-robust CI on contaminated data, reflecting the estimator's stability.
- Specialised packages: `robustbase`, `WRS2` (Wilcox), `quantreg` (quantile regression), `MASS::rlm`.

For Reviewers

What to look for in a paper using this method.

- **Common misapplications.**
- **Diagnostics that should be reported but often aren't.**

- **Red flags in tables and figures.**
- **What to verify.**
- **What an adequate Methods paragraph must contain.**

See also — labs in this chapter

- Descriptive statistics and Table 1